Intermediate Band Solar Cells: Significance of Intermediate Band Mobility

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Intermediate band solar cells (IBSCs) present an advantageous distinction from the conventional p-n junction solar cell, surpassing the Shockley-Queisser (SQ) efficiency limit [1]. The carrier mobility in the intermediate band (IB) has the potential to be a significant constraint in achieving optimal efficiency. Since there is no current produced directly by the IB, the necessity of carrier mobility in the IB is unclear at first. To achieve optimal efficiency, the carriers excited into the IB from the VB must be equal to the carriers excited from the IB to the CB. The subgap excitation rate can vary greatly in different locations, providing the need for a transport method to equalize carrier densities. Consistent with previous research, this report exemplifies the situations where the influence of IB mobility significantly impacts the efficiency of the solar cell.

I. INTRODUCTION

A. Efficiency Limit

The SQ limit prescribes a maximum efficiency of 31% for single p-n junction solar cells under one-sun unconcentrated blackbody radiation from an object at 6000K, (approximate temperature of the sun) and 40.7% under fully concentrated blackbody radiation [1]. These limitations have since been refined, but with similar values. This limit is a common benchmark for technologies aimed to rival conventional solar cell efficiency. Seeking to surpass this efficiency, IBSCs introduce a third energy band between the valence and conduction band of the solar cell material. This allows for the absorption of lower-energy photons without compromising the open-circuit voltage. A detailed balance analysis of IBSCs, conducted by Luque and Marti (LM), limits the achievable efficiency of an IBSC to 46.8% under unconcentrated blackbody radiation and 63.2% under fully concentrated blackbody radiation [2]. IBSCs present an efficiency which significantly surpasses the SQ limit, potentially revolutionizing the solar cell industry.

Detailed balance limits make significant approximations and idealizations to achieve a general efficiency limit. The LM limit assumes perfect absorption, purely radiative recombination, and infinite carrier mobility within the material [2]. In physical scenarios, there are several recombination mechanisms at play, the absorption of each region varies significantly, and the mobility of carriers can be far less than ideal. The effects these parameters have on the efficiency of IBSCs are not well understood and have the potential to hinder the practical implementation of IBSCs if neglected when engineering the IBSC materials [9].

B. Intermediate Band Mobility

As no carriers are directly extracted from the intermediate band, electrons excited to the intermediate band must then be excited to the CB in order to contribute to the power generation of the solar cell. Within the intermediate band electrons have the potential to drift or diffuse to other areas of the intermediate band but are limited by the carrier mobility of the band.

In some cases, the generation rate from the VB to the IB is larger than the generation rate from IB to CB, or vice versa. This discrepancy in generation rate is known as a mismatch and can occur both locally within certain areas of the band and globally across the IB. In the case of local mismatch, the electrons must be able to move through the material to other locations to balance local transitions. The research conducted within this report will discern when the mobility of carriers within the intermediate band has a substantive effect on the achievable efficiency of an IBSC.

C. Simudo

The IBSC band structure seen in Fig. 3 presents a complex dynamic that can be difficult to resolve. The depletion regions at the boundaries within the cell are governed by the drift-diffusion equation and Poisson's equation, leading to a difficult differential equation that must be solved numerically [7].

To model this complex system, the python package Simudo facilitates the simulation of an IBSC using the finite element method to solve the coupled Poisson and drift-diffusion equations throughout the device [7]. This model also incorporates Shockley-Read-Hall (SRH) recombination, surface recombination, and trapping. These recombination methods are essential to the

study of intermediate band mobility as they present one of the largest issues in the practical efficiency of IBSCs. Through the Simudo software, the absorption coefficients, filling fraction and carrier mobility in each region are also spatially variable parameters. These can each be modified to observe their impacts.

II. THEORY

A. PN Junction

Before delving into the complexities of IBSCs, an understanding of the limitations of conventional solar cells can be very useful. Beginning with a generalized semiconductor material consisting of an atom with a valency of four, each atom forms a covalent bond with four other atoms in a perfect crystal. Note that there are numerous combinations of atoms with different valencies that can be used, but valency four atoms are the more common.

The material's crystal structure has two main energy bands, the valence band (VB) and conduction band (CB) separated by a band-gap E_g. Each band has a number of allowed discrete states that electrons can occupy. The number of these states per unit volume is known as the density of states. The number of electrons within each band at a given temperature is given by the Fermi-Dirac distribution [3]:

$$f(E) = \frac{1}{e^{(E-E_F)/kT} + 1}. \tag{1}$$

Here, E is the energy of an allowed electron state, k is Boltzmann's constant, T is the temperature of the material and E_F is the Fermi energy. The Fermi energy is defined as the energy at which the probability of occupancy is 0.5. For an intrinsic semiconductor with no impurities at room temperature, the Fermi energy is located halfway between the VB and CB.

If an atom with a valency of five is introduced into this crystal, then it will produce an additional electron that will not form a bond and will act as a nearly-free electron within the CB, loosely bound by the nucleus of the impurity. Similarly, if an atom of valency three is introduced, then it will leave an additional vacancy in the VB near the impurity site [3]. In semiconductor physics, the empty space left by an excited electron within the VB is known as a hole. Although holes are not physical particles, they can be treated as such, with an effective mass and charge assigned to them. The introduction of holes greatly simplifies the conceptualization of semiconductor physics.

Semiconductors are doped with these two types of impurities to form what are known as n-type and p-type materials, so named because of the additional negative and positive carriers respectively [3]. An example of impurities, also known as dopants, being introduced into a silicon lattice is seen in Fig. 1. Boron provides the p-type dopant and Phosphorus provides an additional electron to introduce n-type doping.

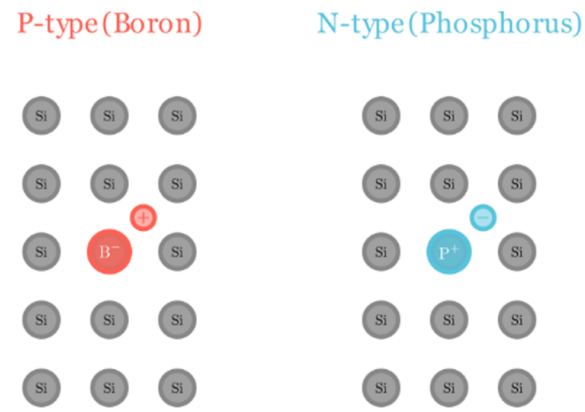

FIG. 1. Illustration of P-type (Boron) and N-type (Phosphorus) doping in Silicon. The p-type doping results in additional holes in the valence band while n-type doping results in additional electrons.

Within these n-type and p-type materials, the Fermi energy of the material is altered locally by the addition of holes or electrons [3]. This shifting is caused by an increase in the number of holes in the VB or electrons in the CB, thus shifting the Fermi-Dirac distribution to center around a new position. This creates a global change in the effective Fermi energy, shifting the effective Fermi energy higher or lower depending on the type of doping. Additional electrons shift the Fermi energy of the material higher while additional holes shift the energy lower.

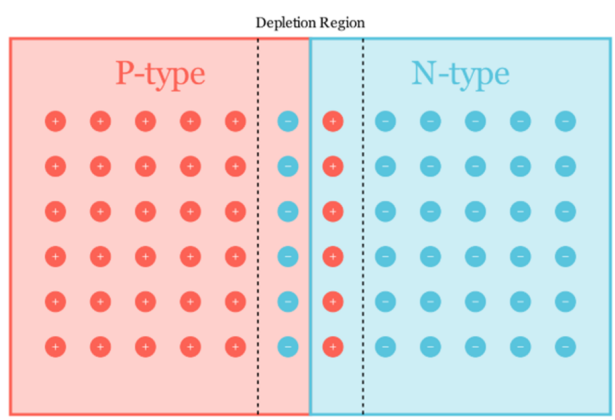

FIG. 2. Illustration of the Depletion Region at a PN Junction. The holes and electrons move across the junction to satisfy the valence bands of each atom, producing ions within the depletion region.

When these two materials are abruptly introduced at some boundary as seen in Fig. 2, they form what is known as a p-n junction [3]. The boundary of the device allows the overabundance of carriers in each material to diffuse into the other and recombine. As the carriers diffuse, they leave behind ionized atoms. The exchange of carriers creates what is known as a depletion region around the boundary, as can be seen in Fig. 2. The ionized atoms within the depletion region introduce a variation in the charge density, producing an electric field as prescribed by the Poisson equation:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}. \tag{2}$$

This electric field opposes the diffusion of carriers and causes them to drift against the diffusion current. On the molecular scale this is a fairly chaotic process, but macroscopically the carriers appear to move at a single velocity known as the drift velocity $\vec{v}_d$ [3]:

$$|\vec{v}_d| = |\mu \vec{E}|. \tag{3}$$

Here μ is the carrier mobility and is dependent on the impurity concentration of the material. The resulting current density of carriers created by this drift is described by the drift-diffusion equations [3]:

$$\vec{J}_n = qn\mu_n \vec{E} + qD(\mu)\frac{dn}{dx}, \quad \vec{J}_p = qp\mu_p \vec{E} - qD(\mu)\frac{dp}{dx}. \tag{4}$$

Here $\vec{J}$ is the current density, q is the fundamental charge, n and p are the carrier concentrations, μ is the carrier mobility for each type of carrier, and $D(\mu)$ is the diffusion coefficient, which is also dependent on the carrier mobility [3]. These equations eventually reach an equilibrium where $\vec{J} = 0$ although the drift and diffusion terms are non-zero. The equations are essential to the operation of a solar cell and provides the foundation for the current produced by a solar cell.

B. Conventional Solar Cell

The cornerstone of solar cell physics is the photovoltaic effect, a close derivative of Einstein's Nobel-winning photoelectric effect. The photovoltaic effect dictates that an incident photon, with an energy greater than the band-gap of a semiconductor material, can interact with and excite an electron from the valence band (VB) to the conduction band (CB) [4]. This will produce what is known as an electron-hole pair as there will be a hole left in the VB as the electron is excited.

When a photon flux is introduced to the p-n junction, the excited electrons are forced towards the n-type side by the electric field, and the holes are forced towards the p-type material [4]. This increase in carriers increases the diffusion current, counteracted by a growing electric field, maintaining the drift-diffusion equation at a steady-state with $\vec{J} = 0$.

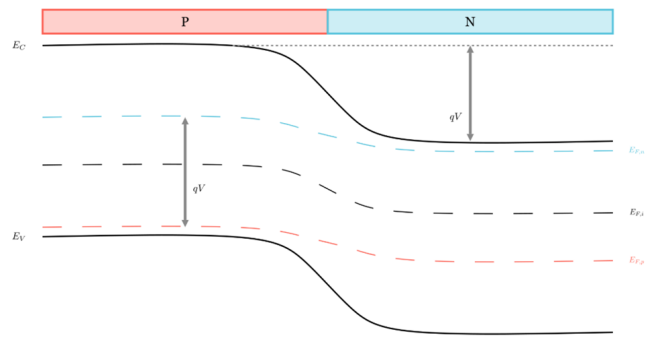

FIG. 3. Band diagram demonstrating the splitting of quasi-Fermi levels ($E_{F,n}$ and $E_{F,p}$) as the solar cell is illuminated.

The increased number of electron-hole pairs creates a difference in the effective Fermi-level seen by the electrons and holes seen in Fig. 3. Conceptually, there will be an increase in holes in the VB, meaning that the Fermi-Dirac distribution must accommodate the increased probability of carriers in the VB. Likewise, electrons in the CB become more populated, leading to a splitting of the Fermi level into two quasi-Fermi levels as can be seen in Fig. 3 with $E_{F,n}$ and $E_{F,p}$ [4].

The splitting of the quasi-Fermi levels creates a voltage potential across the depletion region which can be measured equivalently by two metrics seen in Fig. 3 [4]. The difference in the CB potential across the depletion region, or the difference in the quasi-Fermi levels given by:

$$qV = E_{F,n} - E_{F,p}. \tag{5}$$

By attaching metallic contacts to each end of the p-n junction, this voltage potential can be harnessed to produce a current. The electrons are allowed to flow through some conductive circuit and recombine with the holes in the p-region. As the resistance of the load is reduced, the current increases, reducing the number of available carriers [4]. This process also reduces the potential across the junction as the quasi-Fermi levels grow closer. The optimal power created by a solar cell is a curated balance of the junction potential and the net current density of the cell.

The number of electron-hole pairs generated within the solar cell is dependent on several factors, the most obvious of which is the photon flux into the solar cell. The generation rate is also dependent on the energy of the incident photons, as only photons with an energy higher than the band-gap will be absorbed [4]. We consider an incident solar flux modelled by the spectrum of a black-

body at the temperature of the sun:

$$E(\lambda, T) = \frac{2hc^2}{\lambda^5(e^{hc/\lambda kT} - 1)}. \tag{6}$$

In a perfectly absorptive solar cell, every photon with energy greater than the band-gap of the semiconductor is absorbed [4]. The generation rate is then described by

$$\dot{N}^{absorbed} = \frac{\pi}{h^3c^2}\int_{E_g}^{\infty} \frac{E^2}{e^{E/kT_{sun}} - 1} dE. \tag{7}$$

This is a crucial limiting factor in solar cell efficiency [2]. There are also several systems by which electrons in the CB recombine with holes in the valence band, limiting the total number of excited electrons [3]. The most prevalent case of recombination is radiative recombination, whereby a photon is emitted by an electron in the CB, and the electron recombines with a hole in the VB. The rate of thermal recombination is described by [2]:

$$\dot{N}^{emitted} = \frac{\pi}{h^3c^2}\int_{E_g}^{\infty} \frac{E^2}{e^{(E-qV)/kT_{cell}} - 1} dE. \tag{8}$$

When considering only radiative recombination, the current density through the solar cell is given by:

$$J = q(\dot{N}_{CV}^{abs} - \dot{N}_{CV}^{emit}), \tag{9}$$

where q is the elementary charge.

C. Detailed Balance

A detailed balance limit is an analysis of the maximum efficiency a solar cell can produce. We consider a conventional solar cell with perfect absorption, only radiative recombination, and infinite carrier mobility. In this model, the voltage and current output of the solar cell each vary with the band-gap energy, which can be manipulated to achieve the maximum efficiency for some incident spectrum of light.

The detailed balance limit of conventional, single junction solar cells, first described by the SQ limit, limits the maximum efficiency of a conventional solar cell to 31% under one-sun radiation and 40.7% under fully concentrated blackbody solar illumination [2].

Beyond the detailed balance limit, there are many other factors that limit the efficiency of the solar cell. There are several other forms of recombination that can significantly reduce the efficiency of a solar cell. Trapping is a common issue, when trapping sites such as impurities are introduced that have local state within the band-gap, they "trap" electrons locally [3]. Surface recombination is caused by large numbers of impurities present at the surface of the material and capture electrons similarly to the trapping mechanism. There are several other forms of recombination, each of which contribute to the total recombination rate and reduce the efficiency of the material.

An additional limitation of a solar cell is the mobility of the material, reduced mobility means that in order to achieve the same drift velocity, a larger electric field is required as prescribed by equation (3). The mobility of the material is dictated by the rate of recombination, along with the filling fraction of the band. If a band is filled with electrons, then the rate of collisions increases, and the drift velocity also decreases.

D. Intermediate Band Solar Cells

To surpass the SQ limit, the structure of the solar cell can be altered to improve the absorption of the solar spectrum. The introduction of an intermediate band (IB) between the conduction band (CB) and valence band (VB) allows the solar cell to absorb lower energy photons without compromising the open circuit (OC) voltage [2]. The band structure prescribed by an IBSC can be seen in Fig. 4, wherein three band energies, E_C, E_I and E_V can be seen.

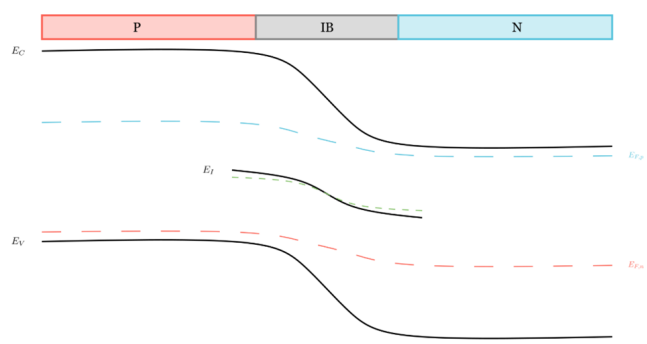

FIG. 4. Band diagram of IBSC structure including the new energy level E_I and quasi-fermi level $E_{F,I}$ introduced.

The conventional solar cell can only absorb photons with an energy greater than its band-gap energy, limiting their efficiency significantly [2]. To circumvent this, IBSCs incorporate an additional band between the CB and VB. This allows electrons in the valence band to be excited to the CB by two consecutive photons, one photon with energy greater than $E_I - E_V$ and a second photon with energy greater than $E_C - (E_I + E_V)$. This allows the solar cell to collect a larger portion of the solar spectrum without compromising the OC voltage.

The current density provided by an IBSC, considering only radiative recombination and perfect absorption, is

calculated with the available photon fluxes [2]:

$$J = q(\dot{N}_{CV}^{absorbed} + \dot{N}_{CI}^{absorbed} - \dot{N}_{CV}^{emitted} - \dot{N}_{CI}^{emitted}) \quad (10)$$

Similarly to the SQ limit, the LM limit is a detailed balance limit of IBSCs, producing a maximum efficiency of 46.8% under unconcentrated sunlight and 63.2% under fully concentrated sunlight [2]. This limit presents a substantive potential for IBSCs to surpass the efficiency of conventional solar cells in practical application.

Creating a detailed balance limit of an IBSC involves balancing both the generation rate and the radiative recombination rate to achieve optimal efficiency. This is done by modifying the band-gap energy, the IB energy and the quasi-Fermi levels to obtain the optimal efficiency. The heatmap presented in Fig. 5, demonstrates the relationship between the maximum efficiency of different combinations of E_I and E_g.

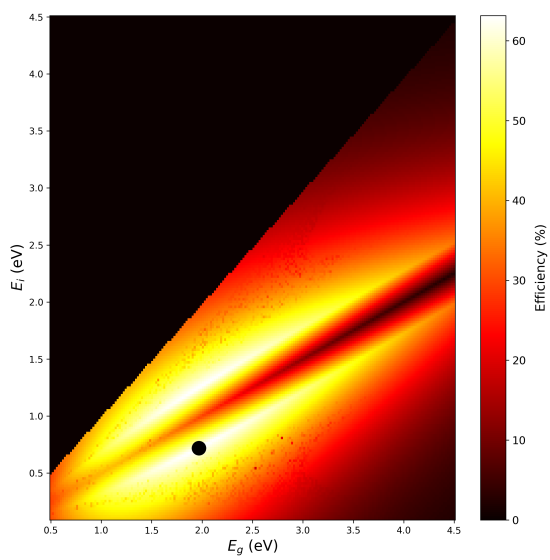

FIG. 5. Heatmap of detailed balance limit for IBSCs as a function of E_I and E_g.

Here the black dot represents the absolute optimal combination of band-gap energy and IB energy with fully concentrated sunlight. This value corresponds to the 63.2% efficiency found by LM in [2].

III. INTERMEDIATE BAND MOBILITY

The LM limit assumed infinite mobility in all bands [2], however, in practical IBSCs, the intermediate band has an increased non-radiative recombination rate because of the additional impurities. The introduction of this recombination provides the basis for carrier mobility influencing how effective the IB process can be. Does low carrier mobility significantly impact efficiency?

Consider the case of an IB located below $\frac{1}{2}E_g$, where E_g is the band-gap depicted in Fig. 6 [9]. If we only allow photons with energy E_{IV} to E_{CI} to be absorbed by the first transition and photons with energy between E_{CI} E_g to be absorbed by the second transition, and we assume perfect absorption then the flux of electrons generated will be:

$$\dot{N}_{IV} = \int_{E_{IV}}^{E_{CI}} \Phi_0(E)dE, \quad \dot{N}_{CI} = \int_{E_{CI}}^{E_g} \Phi_0(E)dE \quad (11)$$

Note that the bounds of integration and subscripts must be flipped for the case where $E_I > \frac{1}{2}E_g$ ($E_{CI} < E_{IV}$):

$$\dot{N}_{CI} = \int_{E_{CI}}^{E_{IV}} \Phi_0(E)dE, \quad \dot{N}_{IV} = \int_{E_{IV}}^{E_g} \Phi_0(E)dE \quad (12)$$

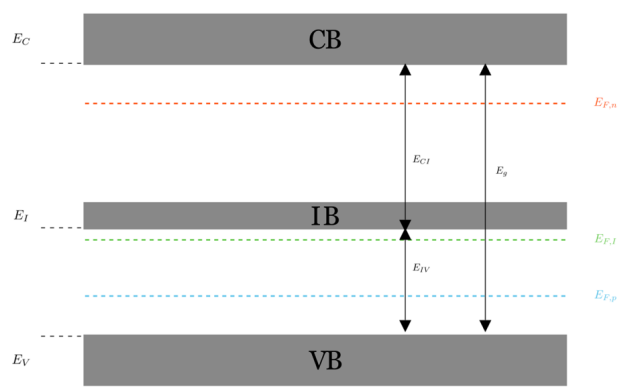

FIG. 6. Illustration of the Intermediate Band Solar Cell band structure showing E_C, E_I, and E_V. Notably the IV subgap energy is lower than the CI subgap energy.

Next, we introduce a parameter to quantify the difference in generation rate between the two transitions. The flux values can then be used to characterize a value deemed the global current mismatch in [9]:

$$m_G = \frac{|\dot{N}_{CI} - \dot{N}_{IV}|}{\dot{N}_{CI} + \dot{N}_{IV}}. \quad (13)$$

This global current mismatch defines how disproportionate the availability of photons is between the two transitions. When the global current mismatch is zero, the solar cell is able to achieve the maximum efficiency at the defined CB and IB energy [9].

Practically speaking, there is no such thing as perfect absorption. Instead, we prescribe each intermediate

band absorption process with an absorption coefficient, α_{IV} and α_{CI} as in [9]. Each of these coefficients will determine the availability of each process to absorb photons. These coefficients will be zero below their respective band-gap energy, and will otherwise follow:

$$\alpha_{IV}(E) = N_I \sigma_{IV}^{opt}(E)(1 - f_I), \tag{14}$$

$$\alpha_{CI}(E) = N_I \sigma_{CI}^{opt}(E) f_I. \tag{15}$$

Here f_I is the filling fraction, a ratio of the electron density to the density of free states within the intermediate band and σ^{opt} is the optical capture cross-section [9].

Current mismatch is not solely a global factor, the absorption coefficients are a function the incident photon flux, which decreases deeper within the cell. Therefore, at a position x within the intermediate band, the local current mismatch can be defined as in [9]:

$$m_L = \frac{\int (g_{IV}(x) - g_{CI}(x))^2 dx}{\int (g_{IV}(x) + g_{CI}(x))^2 dx}, \tag{16}$$

Where $g(x)$ is the net flux of electrons for each process. At each location x within the intermediate band, the generation rate is defined by [9]:

$$g_{IV}(x) = \int_{E_{IV}}^{E_{CI}} \Phi(x, E)\alpha(x, E) dE. \tag{17}$$

Here is where the mobility of the intermediate band becomes crucial. If the global current mismatch is minimal, then the two processes should absorb an equal amount of photons, however if there is a local mismatch within the IB, there must be a mechanism by which the electrons can be transported to the location of the local mismatch from a different location to balance the transitions [9]. If the mobility is too small within the IB, then the electrons will recombine before they are able to move towards or away from the local mismatch. This inability to satisfy one of the transitions at a certain location within the IB will reduce the efficiency of the solar cell by limiting the total carrier flux supplied by the IB.

IV. PREVIOUS RESEARCH

Research into the effects of intermediate band mobility have generally reached similar conclusions, demonstrating the relevance of the IB mobility under certain criteria.

First, in Simudo: a device model for intermediate band solar cells [7], the Simudo software is used to investigate the effects of low intermediate band mobility. The results demonstrate that when the optical absorption is well matched in both the valence to intermediate (IV) process and the intermediate to conduction (CI) process, the efficiency of the device is relatively unaffected by any variation in IB mobility. However, when the optical absorptions of the two processes are not well balanced, the intermediate band mobility can have a significant impact on the efficiency of the device. Notably this simulation was performed with a constant mobility model, meaning the carrier mobility within the IB remains constant.

To expand on this observation, the research performed in Intermediate band mobility requirements in intermediate band solar cells [9] better demonstrates the broad effects of the IB mobility; specifically as they relate to the filling fraction of the IB. The results demonstrated in [9] show that as the IB mobility is most influential when the optical absorptions of the IV and CI processes are mismatched.

The results of these simulations are each limited to a single band-gap energy $E_g = 1.67eV$. To better comprehend the universal effects of the IB mobility, the effects must be demonstrated at alternate band-gap energies.

The results also do not incorporate filling fraction dependent model for the IB mobility, but instead treat it as a constant. The Simudo software has since been updated to provide a basic model for the mobility in the IB. IB mobility is modelled by:

$$\mu = \mu_0 \left[(1 - f_I) - \frac{1}{2} \left(f_I^{\frac{1}{3}} (1 - f_I)^{\frac{4}{3}} \right) \right]. \tag{18}$$

Here the parameter μ_0 is the baseline mobility specified for the material and is used to modify the amplitude of the mobility function. The equation provides a general expected form of the mobility within a sharp energy level, as is used during Simudo simulations, and will decrease as the intermediate band fills or depletes.

V. METHODOLOGY

This study aims to expand on the results derived from the previous research. To fully demonstrate the dynamic effects of IB mobility, the effects must be studied under alternative conditions. Specifically, the simulations in the aforementioned studies only simulate one band-gap energy. To analyze the effects of the IB mobility more comprehensively, a new band-gap energy is selected to ensure that the effects of intermediate band mobility are not unique to a single band-gap energy $E_g = 1.67eV$. The new band-gap energy is chosen at $E_g = 2.5eV$, as can be seen in Fig. 7. This band-gap energy is located on the opposite side of the optimal band-gap energy, $E_g = 1.93$ eV, to determine if the effects of intermediate band mobility (μ_I) are maintained at a different band-gap energy.

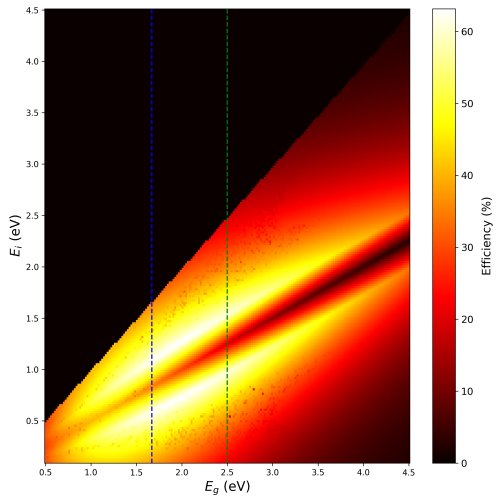

FIG. 7. Heatmap of detailed balance limit for IBSCs with the band-gap energies used for simulation. $E_g = 1.67$ eV shown in blue and $E_g = 2.5$ eV shown in green.

Before delving into new simulation data, the data accrued by previous research was verified to ensure both new and previous results would concur. Several simulations were run to attempt to achieve the same results. However, these led to small discrepancies, on the order of 1%, between new and old results. The source of this error was assessed and traced to a small factor of π in the permeability of free space calculation within Simudo, caused by a previous python package issue.

Once this issue was resolved, the simulation of the solar cell was set up. The parameters seen in Table I were used as the constants to define the simulated solar cell. The parameters are chosen to replicate the IBSCs found in [8], [9], and [10] which demonstrate a typical quantum dot based IBSC.

TABLE I. Simulation Constants

Value	Definition
$X = 1000$	Solar concentration
$\sigma_{CI}^{th} = \sigma_{IV}^{th} = 1.2 \times 10^{-15} cm^2$	Thermal cross-section
$v_{CB}^{th} = v_{IV}^{th} = 2 \times 10^7$	Thermal velocity
$N_I = 1 \times 10^{17} cm^{-3}$	IB density of states
$N_c = N_v = 5 \times 10^{18} cm^{-3}$	CB and VB density of states
$\mu_n = \mu_p = 2000 cm^2 V^{-1} s^{-1}$	Carrier mobility
$\alpha_{cv} = 1 \times 10^4 cm^{-1}$	CV absorption coeff.
$\epsilon_r = 13$	Relative permittivity
$T = 300\ K$	Temp. of solar cell
$T_{sun} = 6000\ K$	Temp. of sun
$w_n = w_p = 1 \mu m$	Thickness of n and p layer
$N_A = N_D = 10^{17} cm^{-3}$	p & n doping conc.
$NFSF = 1 \times 10^{19} cm^{-3}$	Front surface field doping conc.
$f_0 = 0.5$	Initial filling fraction of IB

With these constant values, the simulations required several different variables to better analyze which combinations cause IB mobility to become a significant factor. First, each of the band-gap energies are simulated with four unique IB energies associated with their respective global mismatch values as defined in (13). These energies are shown in Table II, along with their global mismatch value. Notably each band-gap energy requires its own set of IB energies in order to maintain the same global mismatch percentage.

TABLE II. Simulation Sub-gap Energies

m_G	$E_I(E_g = 1.67\ eV)$	$E_I(Eg = 2.5\ eV)$
0%	0.58975 eV	0.97391 eV
10%	0.55447 eV	0.93272 eV
25%	0.49547 eV	0.86227 eV
50%	0.37576 eV	0.71070 eV

Each IB energy is also set so that $E_I < \frac{1}{2} E_g$ Along with these varying band-gap energies, three other parameters are varied: the IB mobility (μ_I) and the optical capture cross-section (σ^{opt}) of the CI and IV processes. These variables are presented in Table III.

TABLE III. Simulation Variables

Variable	Values
$\mu_I (cm^2/Vs)$	0.00001, 0.001, 1, 30, 100
$\sigma_{IV}^{opt} (cm^2)$	1×10^{-13}, 5×10^{-13}
$\sigma_{CI}^{opt} (cm^2)$	1×10^{-13}, 5×10^{-13}

Each combination of these variables is used in an individual simulation using the Simudo software. Each simulation alters the thickness of the IB material within the device to find the optimal efficiency.

VI. RESULTS & DISCUSSION

A. Correlated Efficiency Trends

With the parameters established, each combination of variables was simulated using the Simudo software. The Simudo simulations were performed using a p-I-n structure and the parameters laid out in I.

The resulting efficiency plots demonstrate an excellent correlation between the results at both band-gap energies. The plots in Fig. 8 and Fig. 9 show the strikingly similar trends in efficiency with respect to the IB mobility in each band-gap energy case. These close correlations are not unique to this combination of σ_{CI}^{opt} and σ_{IV}^{opt}, but appear in every efficiency plot, as can be seen in the Appendix in Fig. 16 and Fig. 17. These strong correlations

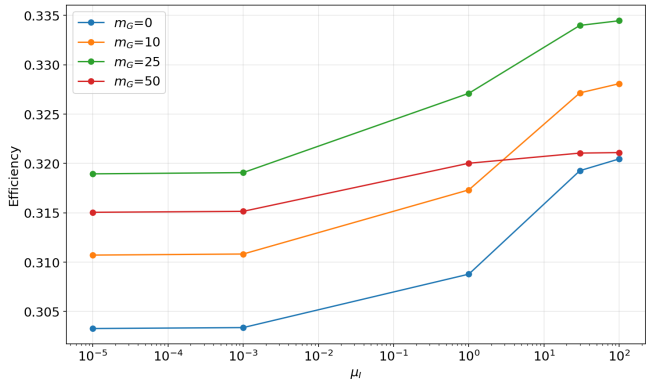

FIG. 8. Efficiency as a function of IB Mobility (μ_I) for mismatched σ^{opt}. Simulated with band-gap $E_g = 1.67$ eV, $\sigma^{opt}_{CI} = 5 \times 10^{-13}$ and $\sigma^{opt}_{IV} = 1 \times 10^{-13}$.

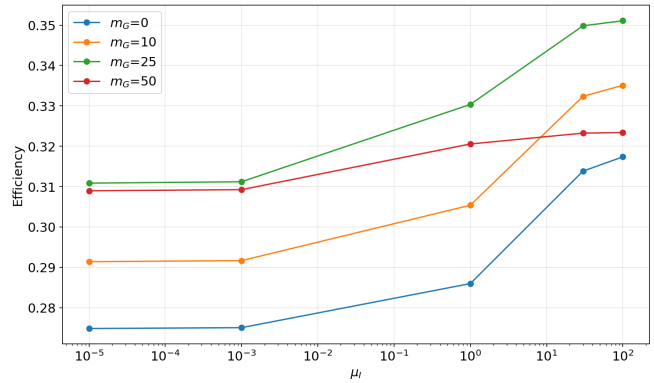

FIG. 9. Efficiency as a function of IB Mobility (μ_I) for mismatched σ^{opt}. Simulated with band-gap $E_g = 2.5$ eV, $\sigma^{opt}_{CI} = 5 \times 10^{-13}$ and $\sigma^{opt}_{IV} = 1 \times 10^{-13}$.

demonstrate that the efficiency trends are not unique to any single band-gap energy, but are present across multiple band-gap energies. The efficiency trends can also be extrapolated to presume that these trends are present at values around the simulated points, shown in Fig. 7.

B. IB Mobility Influence

With these trends present at multiple band-gap energies, a deeper investigation into the cause of these trends can shed light on the broader relevance of IB mobility rather than a potentially unique case. The plots illustrated in Fig. 8 and 9 also present the case described in previous research as situations where the IB mobility is influential. This is caused by a large difference in the σ^{opt} of the IV and CI processes.

In this case, the σ^{opt}_{CI} is five times larger than σ^{opt}_{IV}. As a result, the absorption coefficient of the CI process defined in (14) will be larger than the absorption coefficient of the IV process defined in (15) for an IB filling fraction of $f = 0.5$.

To better visualize the effects of this difference, Fig. 10 presents the IB filling fraction as a function of location within the IB material. The incident photons enter from the left of the plot, which is also the P-type to IB material junction. The plot demonstrates the expected layout of an IB filling fraction plot when the σ^{opt}_{CI} and σ^{opt}_{IV} are matched and the effects of IB mobility are negligible. The depletion regions on either end of the plot are evident in the filling fraction, and the majority of the plot has a filling fraction $f = 0.5$.

To contrast, the plot displayed in Fig. 11 is the filling fraction of the IB taken at short circuit with the same σ^{opt} ratio seen in Fig. 8 and Fig. 9. This plot exemplifies the issue with optical absorption discrepancy, most noticeably on the left of the plot where the cell is illuminated from. The plots all have a sharp depletion of electrons caused by the difference in the two processes' absorption coefficients.

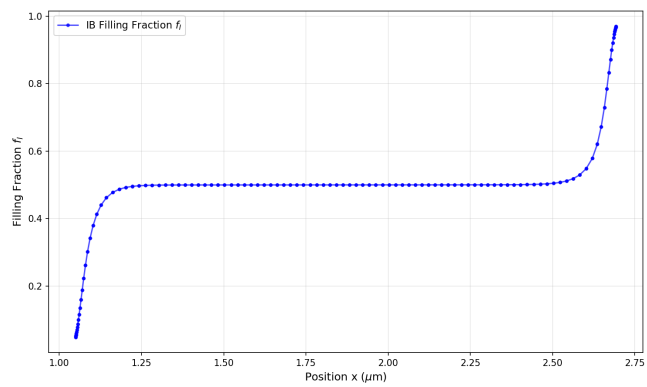

FIG. 10. Optimal IB filling fraction at short circuit with matched σ^{opt}. $E_g = 1.67$ eV, $\sigma^{opt}_{CI} = 1 \times 10^{-13}$ and $\sigma^{opt}_{IV} = 1 \times 10^{-13}$.

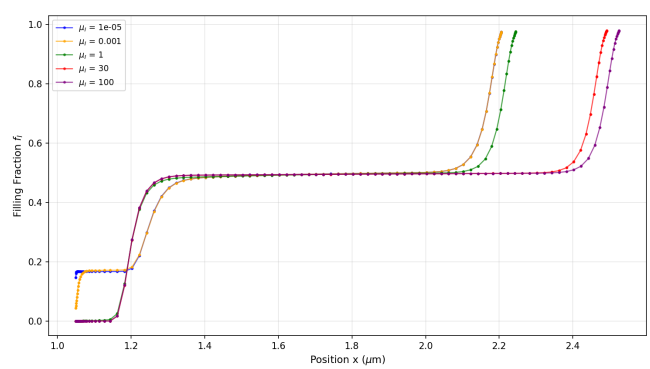

FIG. 11. IB filling fraction at short circuit with mismatched σ^{opt}. $E_g = 1.67$ eV, $\sigma^{opt}_{CI} = 5 \times 10^{-13}$ and $\sigma^{opt}_{IV} = 1 \times 10^{-13}$.

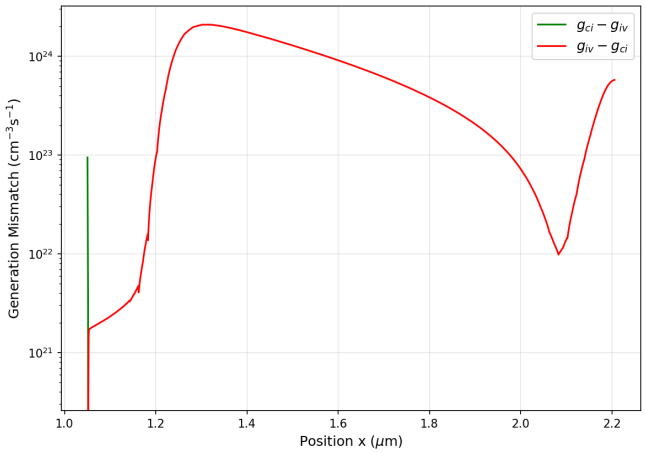

FIG. 12. Optical generation difference between IV and CI processes at short circuit with low IB mobility $\mu_I = 1 \times 10^{-5}$. Simulated with $E_g = 1.67$ eV, $\sigma_{CI}^{opt} = 5 \times 10^{-13}$ and $\sigma_{IV}^{opt} = 1 \times 10^{-13}$.

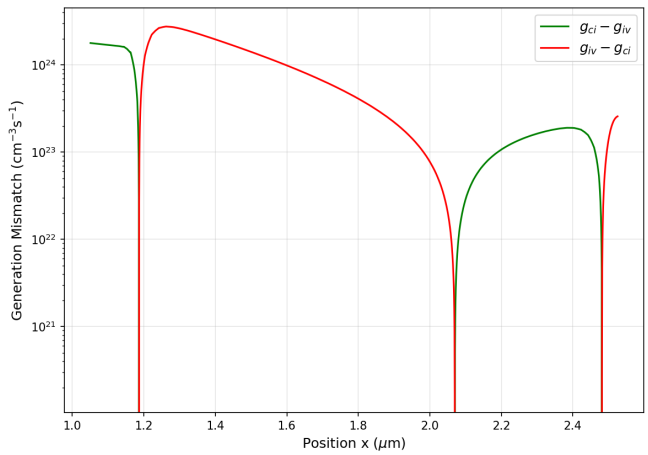

FIG. 13. Optical generation difference between IV and CI processes at short circuit with high IB mobility $\mu_I = 1 \times 10^{2}$. Simulated with $E_g = 1.67$ eV, $\sigma_{CI}^{opt} = 5 \times 10^{-13}$ and $\sigma_{IV}^{opt} = 1 \times 10^{-13}$.

Notably, as the IB mobility increases, the number of electrons on the left of the plot increases to $f = 1/6$, as is also noted in [9], where the subgap generations are balanced by the filling fraction.

This shift in filling fraction demonstrates the effect of the reduced IB mobility as the electrons excited to the IB are not swept away by the electric field as well as in the high μ_I case. This increased recombination reduces the efficiency of the IBSC, as is demonstrated by the results in Fig. 8 and Fig. 9. These results support the previous results found in [7] and [9], with a positive trend in efficiency as the IB mobility increases in the case of mismatched σ^{opt}. Although notably the filling fraction plots shown in [9] appear very different because of the constant μ_I model used.

To corroborate the effects of this shift in filling fraction, the difference in optical generation shown in Fig. 12 and Fig. 13 are taken at short circuit for $\mu_I = 1 \times 10^{-5}$ and $\mu_I = 1 \times 10^{2}$ respectively. These plots present the difference between the CI and IV optical generation rates throughout the IB material. These plots clearly demonstrate that as μ_I increases, the CI process begins to dominate around the P-type junction, leading to a depletion of electrons in the filling fraction plots. The dominance of the IV process at low μ_I points to the cause of the reduced efficiency. Reduced CI transitions lead to an overall reduction in the efficiency of the device.

Overall, the results of this report are consistent with efficiency trends presented in previous research, demonstrating that with large differences in σ_{opt} for the CI and IV processes, IB mobility plays an influential role in determining the efficiency of the device. The results show that this influence is not unique to a single band-gap energy, but is well correlated to other band-gap energies with similar influence.

There were some efficiency trends in the results which were not present in previous studies, and should be investigated to determine if the results are the result of calculation errors or if the results demonstrate a new, unexplored impact of IB mobility.

C. Future Work

The results of the simulations conducted in this report yielded some unexpected results, specifically in the case where σ^{opt} is matched for the two processes. The results in [9] demonstrate a mainly independent trend from any effect caused by the IB mobility in this case. However, during the simulations conducted in this study, there were notable negative trends in the efficiency of the IBSC as μ_I increased.

The plot in Fig. 14 demonstrates the expected, unaffected trend in efficiency with small fluctuations that can be attributed to error in the optimization process. Fig. 15 presents a much larger shift in efficiency which cannot be accounted for in the optimization error.

To further understand the effects of IB mobility across every band-gap energy, these same simulations can be performed with several different parameters changed. The band-gap energies could be closer to or farther from the optimal value, the band structure could instead have $E_I > E_g/2$, or the entire physical structure of the device could be altered to a p-I-n structure rather than n-I-p.

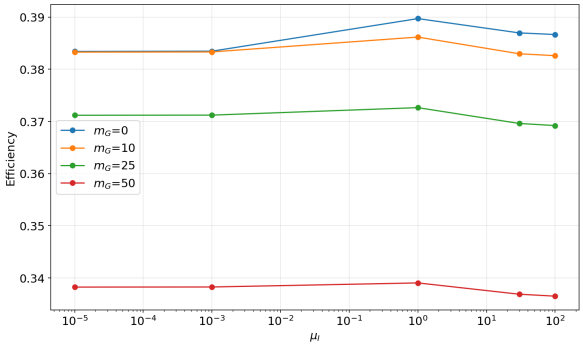

FIG. 14. Efficiency as a function of IB Mobility (μ_I) for matched σ^{opt}. Simulated with $\sigma_{CI}^{opt} = 5 \times 10^{-13}$ and $\sigma_{IV}^{opt} = 5 \times 10^{-13}$.

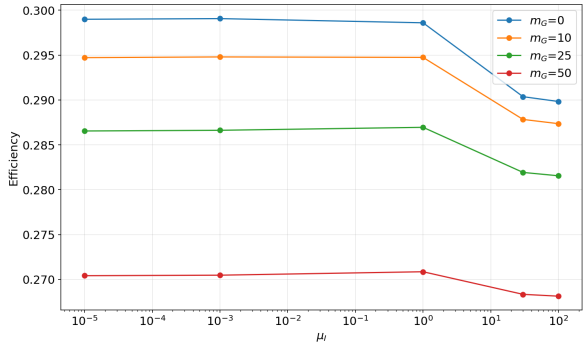

FIG. 15. Efficiency as a function of IB Mobility (μ_I) for matched σ^{opt}. Simulated with $\sigma_{CI}^{opt} = 1 \times 10^{-13}$ and $\sigma_{IV}^{opt} = 1 \times 10^{-13}$.

VII. CONCLUSION

The results presented within this report demonstrate the complex nature of understanding the importance of various parameters within IBSCs. The results demonstrate a clear correlation between the efficiency trends of multiple band-gap energies, proving that these trends are not unique to any particular band-gap energy. The results do however present unexpected trends in efficiency that require further investigation to determine the exact cause, and determine whether the trends are significant.

ACKNOWLEDGMENTS

I would like to sincerely thank Dr. Jacob Krich for sharing his expertise on the subject with a mastery and grace I will never fully comprehend. His explanation of complex topics such as this one is beyond wonder. I would also like to thank the other creators of Simudo, Eduard Dumitrescu and Matthew Wilkins, without their brilliantly designed framework and well commented code I would still be staring at an impressive and hopeless selection of errors.

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APPENDIX

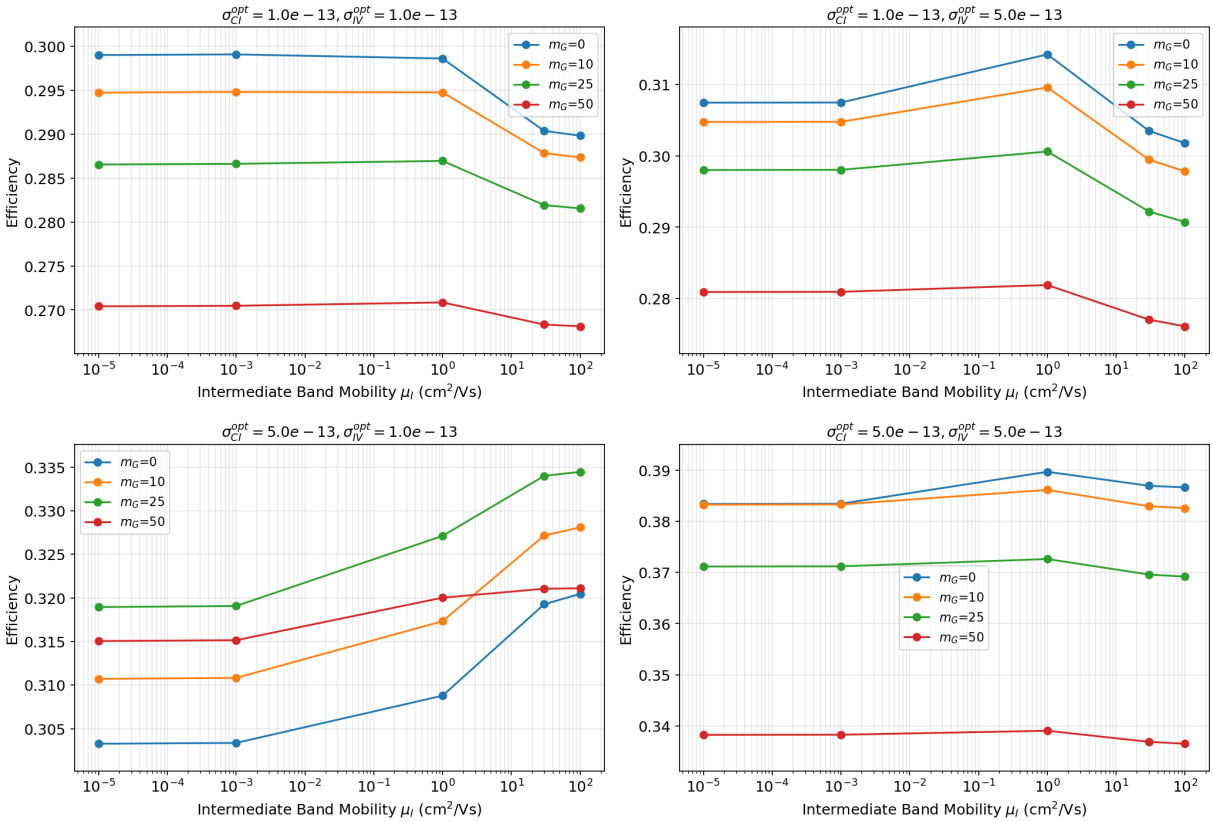

FIG. 16. Efficiency as a function of IB Mobility (μ_I) with band-gap $E_g = 1.67$ eV.

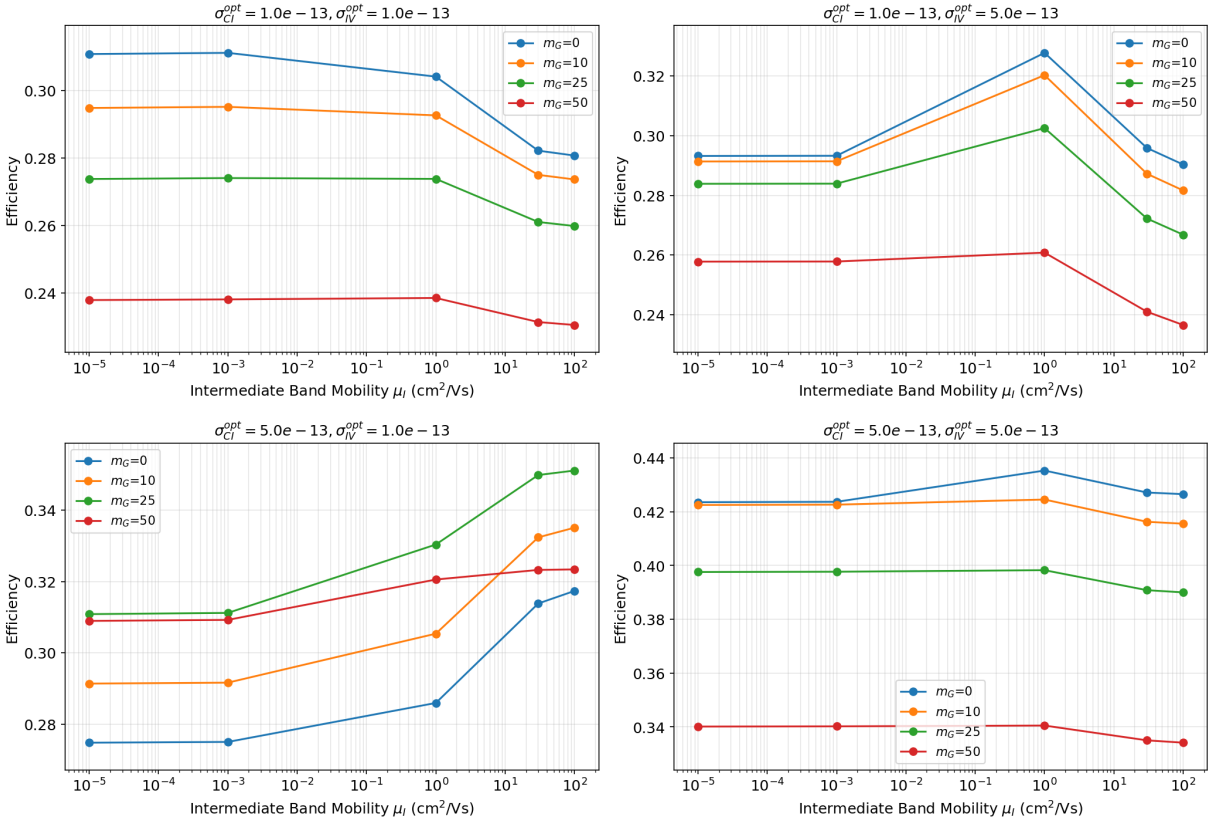

FIG. 17. Efficiency as a function of IB Mobility (μ_I) with band-gap $E_g = 2.5$ eV.